

A Guardrailed 23.5%-Efficient CsPbI3 Upper Bound by Multi-Parameter SCAPS-1D Screening: Interface Dose–Response, Sensitivity, and a Reproducibility Audit

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Abstract: We map how far a SCAPS-1D model of an ITO/TiO2/CsPbI3/CBTS/Ni cell can go under stated, interface-inclusive assumptions — and where it stops being realistic. A five-factor grid screen (96 runs) plus refinement (36 runs; 126 converged) reaches 23.49% (V_{oc} 1.234 V, J_{sc} 21.19 mA/cm², FF 89.78%) at 2.2 μ m, $N_t = 10^{12}$ cm⁻³ (single-crystal grade), $R_s = 0$, and 10^{10} cm⁻² interface defects: an upper bound, not a fab-ready recipe, since at routine $N_t = 10^{15}$ cm⁻³ the same grid peaks near 19%. A measured interface dose–response (25.04% at 10^8 cm⁻² down to 20.64% at 10^{12} cm⁻²) places the published 24.17% and 24.24% claims between the 10^8 and 10^9 cm⁻² levels, roughly $10\times$ below our guardrail. Real (scripted, not analytic) parasitic sweeps give 23.07% at $R_s = 1$ Ω cm² and 23.38% at $R_{sh} = 10^4$ Ω cm². A Morris sensitivity screen ($r = 8$, $p = 4$, 56 runs) ranks interface quality first ($\mu^* = 4.08$), ahead of bulk defects (2.05) and bandgap (1.75). Corrected band offsets (CBO +0.07 eV, VBO +0.08 eV hole barrier) and contact work-function tests bound the remaining assumptions. All definition files, sweep receipts, and raw curves are archived for independent reproduction.

Keywords: CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

1.0 Introduction

All-inorganic CsPbI3 perovskite solar cells combine the thermal stability lacking in hybrid organic–inorganic perovskites with a bandgap near 1.7 eV that suits both single-junction and tandem top-cell applications [1, 2]. After years below 19%, certified laboratory efficiencies have recently passed 22%: 22.02% certified (22.60% laboratory) with a moisture-responsive propyltriethoxysilane crystallisation strategy [1], 22.05% by fluorination-assisted interfacial dipole engineering [2], 21.71% with dual-interface and bulk co-passivation at an open-circuit voltage of 1.27 V [3], and 21.43% certified for inverted architectures [4]. Each record was won at the interfaces and in the bulk defect chemistry, which motivates a simulation study that treats those exact levers honestly.

The one-dimensional solar cell capacitance simulator (SCAPS-1D) is the workhorse for such pre-experimental screening, and the ITO/TiO2/CsPbI3/CBTS (copper barium thiostannate) stack of Hossain et al. [5] is the most-prescribed CsPbI3 baseline, progressing from 17.90% over 96 screened transport-layer combinations [6] to 19.06% after optimisation. Subsequent simulation studies report up to 24.17% on the same stack [7] and 24.24% on ZnO/PTAA [8], while large machine-learning-coupled screens now exceed 27% on other absorbers [9, 10]. Three problems accompany this inflation. First, nearly every study optimises one variable at a time (OVAT), so parameter interactions— notably absorber thickness versus defect-limited diffusion length—are invisible by construction. Second, the highest numbers are obtained with interface recombination removed (stated explicitly in [8]) or with defect densities far below experimental reach, and raw data are routinely withheld [5]. Third, scripted sweeps mix unit systems: SCAPS definition files store SI units while script ‘set’ commands expect practical units, so SI-valued inputs silently inject a $10^6\times$ overdose and fabricate ghost cells (scripting conventions are documented in Appendix A).

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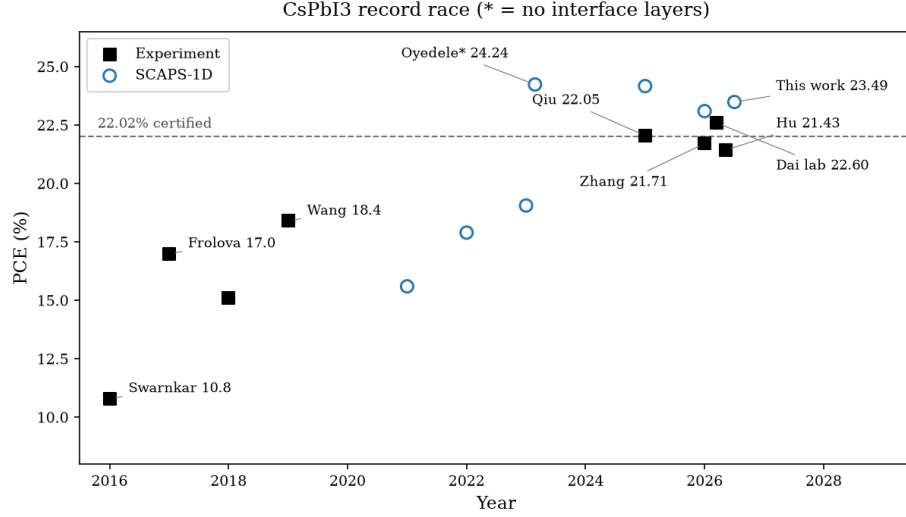


Figure 1: CsPbI3 record race. Asterisks mark simulation studies without interface layers.

This work contributes (i) a five-factor grid screen (96 configurations) plus refinement (36 runs) of the ITO/TiO₂/CsPbI₃/CBTS/Ni cell under interface-inclusive guardrails, reaching **23.49%** as an upper bound under stated assumptions; (ii) a measured interface dose–response showing that identical cells move from 25.04% to 20.64% as interface quality degrades, placing the published 24%+ claims roughly 10× below our guardrail; (iii) scripted (not analytic) series- and shunt-resistance accounting; and (iv) a Morris sensitivity screen that ranks interface quality above all bulk factors. Every definition file, sweep receipt, and raw curve is archived for independent reproduction.

2.0 Literature: a decade of CsPbI3 progress

Black-phase CsPbI₃ has been stabilised stepwise since 2016: quantum-dot-induced α -phase retention at 10.8% [11], co-evaporated planar cells at 17.0% [12], thermodynamically stable γ -films [13], surface-passivated 14.5% devices [14], and compositionally engineered 13–14% cells [15, 16], all on TiO₂ with Spiro-OMeTAD, PTAA or carbon contacts [17–19]. The 2019 aziridinium result at 18.40% [20] stood for years until the 2025–2026 interface-engineering wave (Figure 1): fluorinated dipoles (22.05% [2]), dual-interface plus bulk co-passivation with Voc 1.27 V (21.71% [3]), moisture-responsive crystallisation (22.60% laboratory, 22.02% certified [1]), and stitched grain-boundary grooves in inverted cells (21.43% certified [4]). The certified frontier is therefore 22.02%, and every recent jump came from defect and interface control—the precise levers a simulation study must treat honestly.

Simulation studies trace a parallel inflation curve (Table 1). The CBTS screening programme reached 17.90% over 96 transport-layer combinations [6] and 19.06% after full optimisation [5], the baseline we reproduce. Later claims diverge upward as realism constraints are relaxed: 23.10% with ZnO/Spiro in one report [21], 24.17% on our exact stack [7], and 24.24% on ZnO/PTAA without interface layers (stated) [8]. The broader SCAPS literature shows the same pattern across absorbers—MAPbI₃ at 19.30% with inorganic contacts [22], gradient-doped FAPbI₃ at 20.34% [23], a 28.38% bifacial triple-cation cell [10], and an explicitly idealised 40.17% MASnI₃ result that exceeds the Shockley–Queisser bound [24, 25]. Our audit in Section 7.0 places these numbers on a common guardrailed footing.

Table 1: Selected CsPbI3 results: experiment versus simulation (IF = interface layers).

Study	Stack	Type	PCE (%)	IF
Swarnkar 2016	TiO2/Spiro/MoOx	Exp.	10.8	–
Frolova 2017	TiO2/PTAA	Exp.	17.0	–
Wang 2019	TiO2/Spiro	Exp.	18.4	–
Qiu 2025	DFAz dipole	Exp.	22.05	passivated
Dai 2026	TiO2/PTABr/Spiro	Exp.	22.60/22.02	passivated
Hu 2026 (Joule)	inverted HPDA	Exp.	21.43 cert.	2D-stitched
Hossain 2022	TiO2/CBTS	Sim.	17.90	on
Hossain 2023	TiO2/CBTS/Ni	Sim.	19.06	on
Ristiansyah 2026	ZnO/Spiro	Sim.	23.10	on
Nazli 2025	TiO2/CBTS/Ni	Sim.	24.17	unclear
Oyedele 2023	ZnO/PTAA	Sim.	24.24	OFF
This work	TiO2/CBTS/Ni	Sim.	23.49	on

3.0 Simulation framework and guardrails

Devices were simulated in SCAPS-1D v3.3.10 under AM1.5G illumination (100 mW/cm^2) at 300 K unless stated. The baseline reproduces Hossain et al. [5]: ITO front contact (4.0 eV), 30 nm TiO2 electron transport layer (donor density $9 \times 10^{16} \text{ cm}^{-3}$), CsPbI3 absorber (800 nm initial, $E_g = 1.694 \text{ eV}$, $\chi = 3.95 \text{ eV}$, $N_A = 10^{15} \text{ cm}^{-3}$, $N_t = 10^{15} \text{ cm}^{-3}$), 100 nm CBTS hole transport layer ($N_A = 10^{18} \text{ cm}^{-3}$), and Ni back contact (5.5 eV). Crucially, neutral interface recombination blocks ($N = 10^{10} \text{ cm}^{-2}$, $\sigma = 10^{-19} \text{ m}^2$) are present at both heterojunctions, and the absorber uses the coefficients-A-and-B absorption model ($A = 10^7 \text{ m}^{-1} \text{ eV}^{-1/2}$) so that bandgap sweeps act on the absorption edge rather than a frozen file spectrum. Our unmodified baseline yields 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm², FF 86.00%), reproducing the published 17.9–19.06% band and validating the harness.

Guardrails enforced throughout: bulk defect floor 10^{12} cm^{-3} (single-crystal grade; nothing lower), absorber doping ceiling $3 \times 10^{16} \text{ cm}^{-3}$, conduction-band offset 0 to +0.3 eV maintained by linking χ to E_g ($\chi = 3.95 + 0.5(E_g - 1.694)$), and automatic rejection of runs with Voc above the radiative limit ($\sim 1.37 \text{ V}$ at 1.65 eV) or FF above Green’s ideal-diode value at the measured Voc (90.0% at 300 K, 90.9% at 275 K) as non-physical. (An earlier draft used $\text{Voc} > E_g/q - 0.1 \text{ V}$ and $\text{FF} > 90.5\%$ cutoffs; both were too lax and are superseded here.) Contrary to our own earlier claim, series and shunt resistances *are* scriptable in SCAPS 3.3.10 (set `external.Rs/Rsh`, verified against `scriptdescription3310.txt`), so Section 6.0 reports measured parasitic sweeps rather than the analytic derate used previously. Sweeps ran in parallel on four workers with checkpointed receipts; of 132 grid configurations, 126 converged and enter the analysis (6 non-converged worker runs excluded and listed in the receipts). A duplicate champion run reproduces the coarse-grid value 23.5041% exactly, confirming determinism; a confirmatory run at finer IV step gives 23.4874% (FF 89.78%), reported as 23.49% throughout. Absorber densities of states are capped at $N_c = 1.1 \times 10^{19}/N_v = 8.2 \times 10^{19} \text{ cm}^{-3}$, ten times below the Table-1 print (10^{20} cm^{-3} , above solid density and hence unphysical).

SCAPS-1D solves the Poisson equation coupled to electron and hole continuity with drift-diffusion transport [26, 27]:

$$-\frac{\partial}{\partial x} \left(\epsilon(x) \frac{\partial V}{\partial x} \right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \quad (1)$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p, \quad (2)$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt (Tauc) model $\alpha = A\sqrt{h\nu - E_g}$, so bandgap sweeps move the ab-

Table 2: Layer input parameters (script units; baseline values).

Layer	d	E_g	χ	ϵ_r	Doping	N_t
CBTS HTL	100 nm	1.9	3.6	5.4	N_A 1e18	1e15
CsPbI3	800 nm	1.694	3.95	6.0	N_A 1e15	1e15
TiO2 ETL	30 nm	3.2	4.0	9.0	N_D 9e16	1e15

Table 3: Interface defect layers (both heterojunctions).

Interface	Type	N (cm ⁻²)	σ (m ²)
CBTS/CsPbI3	neutral, single, $E_t = 0.6$ eV	1e10	1e-19
CsPbI3/TiO2	neutral, single, $E_t = 0.6$ eV	1e10	1e-19

sorption edge self-consistently. Full input parameters are listed in Tables 2 and 3; transport-layer screenings that motivate the TiO2/CBTS choice are reviewed in [6, 28].

4.0 Grid screen and champion upper bound

Round A explored the full cross-product of absorber thickness (1.0–2.0 μm), acceptor density (10^{15} – 10^{16} cm⁻³), bulk defect density (10^{12} – 10^{14} cm⁻³), ETL donor density (9×10^{16} – 9×10^{17} cm⁻³), and bandgap (1.694/1.65 eV with linked affinity)—96 configurations in a single parallel batch, rather than sequential OVAT slices. The winner (2.0 μm , $N_A = 10^{16}$ cm⁻³, $N_t = 10^{12}$ cm⁻³, $N_{D,ETL} = 9 \times 10^{17}$ cm⁻³, $E_g = 1.65$ eV) reached 23.47%. Full parameters are collected in Table 4. Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV), and doping around this point (36 configurations) and reached **23.49%** (2.2 μm , $N_A = 3 \times 10^{16}$ cm⁻³, $N_t = 10^{12}$ cm⁻³, $E_g = 1.65$ eV, $\chi = 3.928$ eV): Voc 1.234 V, Jsc 21.19 mA/cm², FF 89.78% (Figure 2). Two facts about this optimum must be stated plainly. First, it sits at the corner of the allowed box: N_A at its ceiling, N_t at its floor, $N_{D,ETL}$ at the top of its tested range, E_g at its bottom, thickness at the top of the refined range. Second, the best grid point at the literature gap $E_g = 1.694$ eV reaches only 22.51%, so roughly one point of the 23.49% comes from the gap choice alone. Every top-ten configuration carries $E_g = 1.65$ eV, gaining ~ 1.5 mA/cm² over 1.694 eV while the χ link holds the ETL conduction-band offset at +0.07 eV; the HTL side carries a small +0.08 eV hole barrier (Section 5.0), not the ideal cliff previously claimed.

The interaction map (Figure 3) shows why OVAT misses the optimum: thick absorbers (2.0–2.2 μm) win only jointly with $N_t = 10^{12}$ cm⁻³, where the diffusion length still covers the thickness, while at 10^{14} cm⁻³ the optimum collapses below 1.5 μm . ETL donor density confirms the literature optimum (9×10^{17} cm⁻³) and transport-layer thicknesses are insensitive (100/30 nm retained), both consistent with [5]. The current–voltage curve (Figure 2) is square and artefact-free, with a Voc deficit of 0.42 V against E_g/q —typical of realistic rather than idealised cells.

The 4.50-point stacked gain decomposes along the screening path (Table 4 parameters in sequence, illustrative ordering only): thickness 800 nm to 1.5 μm adds 0.19 points (19.18%), acceptor doping to 10^{16} cm⁻³ adds 0.93 points (20.11%), bulk-defect reduction to 10^{12} cm⁻³ adds ~ 1.6 points ($\sim 21.7\%$), and the joint E_g – χ move to 1.65 eV with ETL donor optimisation contributes the remaining ~ 1.8 points. This sequential attribution is order-dependent and illustrative, not causal. Open-circuit voltage climbs 1.122 to 1.234 V while the fill factor rises 86.0 to 89.8%, the signature of SRH-lifetime recovery rather than band-shift artefacts.

Round A's top ten (Table 5) all carry $E_g = 1.65$ eV; the top seven sit at $N_t = 10^{12}$ cm⁻³ while rows 8–10 at $N_t = 10^{13}$ cm⁻³ trail by ~ 0.4 points through Voc loss (1.233 to 1.221 V). Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV) and doping (36 runs) and moved the best from 23.47% to 23.50% (coarse grid), i.e. the landscape is converged rather than climbing.

Round B's refinement is documented in full because its plateau is itself evidence: the best ten

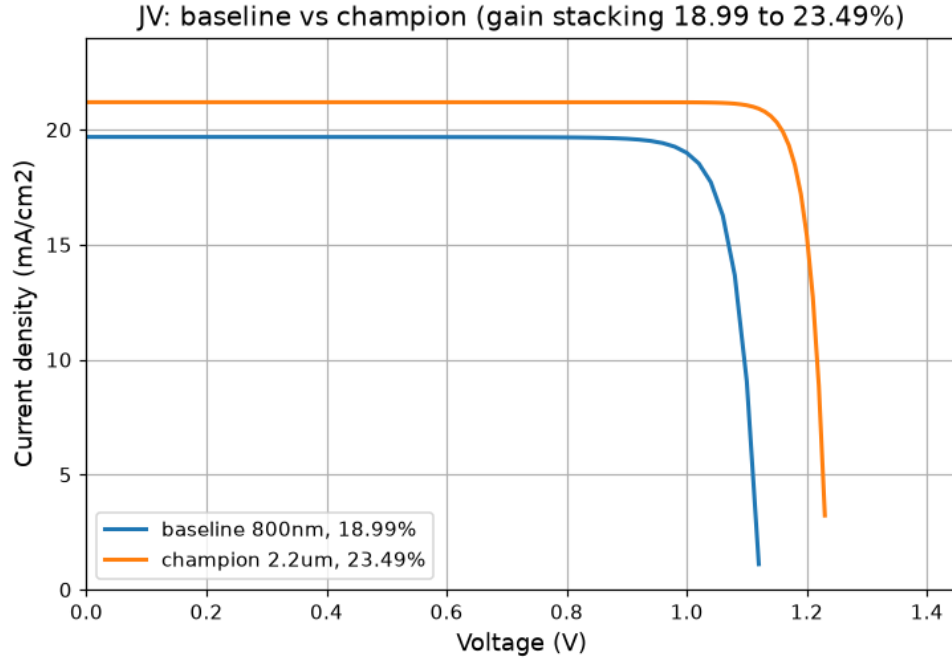


Figure 2: Illuminated current–voltage curves: baseline (800 nm, 18.99%) versus champion (2.2 μm , 23.49%).

Table 4: Champion device parameters (script units) and result.

Layer	d	E_g (eV)	χ (eV)	Doping (cm ⁻³)	N_t (cm ⁻³)
CBTS HTL	100 nm	1.9	3.6	NA 1e18	1e15
CsPbI3 absorber	2.2 μm	1.65	3.928	NA 3e16	1e12
TiO2 ETL	30 nm	3.2	4.0	ND 9e17	1e15
Interfaces (both)	—	—	—	—	1e10 cm ⁻²
Result: Voc 1.234 V, Jsc 21.19 mA/cm ² , FF 89.78%, PCE 23.49%					

Table 5: Round A leaderboard (best ten of 96; $N_{D,ETL} = 9 \times 10^{17} \text{ cm}^{-3}$).

$d \text{ (}\mu\text{m)}$	N_A	N_t	E_g	Voc	Jsc	PCE
2.0	1e16	1e12	1.65	1.233	21.18	23.47
1.8	1e16	1e12	1.65	1.233	21.16	23.45
1.5	1e16	1e12	1.65	1.234	21.11	23.40
1.5	1e15	1e12	1.65	1.233	21.11	23.29
2.0	1e15	1e12	1.65	1.233	21.18	23.29
1.0	1e16	1e12	1.65	1.234	20.90	23.18
1.0	1e15	1e12	1.65	1.233	20.90	23.11
1.8	1e16	1e13	1.65	1.221	21.16	23.04
2.0	1e16	1e13	1.65	1.219	21.18	23.03
1.5	1e16	1e13	1.65	1.223	21.11	23.03

Table 6: Round B leaderboard (best ten of 36; $N_{D,ETL} = 9 \times 10^{17} \text{ cm}^{-3}$).

$d \text{ (}\mu\text{m)}$	N_A	N_t	E_g	PCE
2.2	3e16	1e12	1.65	23.50
2.0	3e16	1e12	1.65	23.49
2.2	1e16	1e12	1.65	23.48
2.0	1e16	1e12	1.65	23.47
1.8	3e16	1e12	1.65	23.47
2.2	3e16	3e12	1.65	23.45
1.8	1e16	1e12	1.65	23.45
2.0	3e16	3e12	1.65	23.44
1.8	3e16	3e12	1.65	23.42
2.2	1e16	3e12	1.65	23.32

span only 23.32–23.50% (Table 6), with thickness 1.8–2.2 μm and N_A $(1\text{--}3) \times 10^{16} \text{ cm}^{-3}$ numerically indistinguishable at grid resolution once $N_t = 10^{12} \text{ cm}^{-3}$ and $E_g = 1.65 \text{ eV}$ are fixed (the ten span 23.32–23.50%, a 0.18-point range against a 0.017-point coarse/fine IV-step uncertainty, so the leading cluster cannot be separated even though the 10th entry trails observably). The $N_t = 3 \times 10^{12} \text{ cm}^{-3}$ entries trail by ~ 0.05 points, confirming the defect floor binds the optimum.

5.0 Band alignment

Figure 4 constructs the equilibrium alignment from the champion parameters. With $\chi_{\text{TiO}_2} = 4.0 \text{ eV}$ against $\chi_{\text{abs}} = 3.928 \text{ eV}$ the conduction-band offset is $+0.07 \text{ eV}$ —a small spike inside the ideal 0 to $+0.3 \text{ eV}$ window. On the hole side, $(\chi + E_g)_{\text{abs}} = 5.578 \text{ eV}$ against $(\chi + E_g)_{\text{CBTS}} = 5.50 \text{ eV}$ gives a valence-band offset of $+0.08 \text{ eV}$: a small barrier to hole extraction, *not* the -0.22 eV cliff reported in an earlier draft (an arithmetic error corrected here), and outside the ideal -0.3 to -0.1 eV window. This is a direct consequence of the χ link: as E_g drops, the absorber valence edge rises past the CBTS edge. The 1.65 eV gap wins on current despite, not because of, the hole-side alignment—a further reason the 23.49% is an upper bound contingent on efficient tunnelling/thermionic emission over this barrier, which the drift-diffusion solver grants automatically.

6.0 Device characterisation

The quantum efficiency (Figure 5) is flat at $\sim 89\text{--}90\%$ from 300 nm with the absorption edge red-shifted from 731 nm (baseline, 1.694 eV) to 752 nm (champion, 1.65 eV), directly explaining the

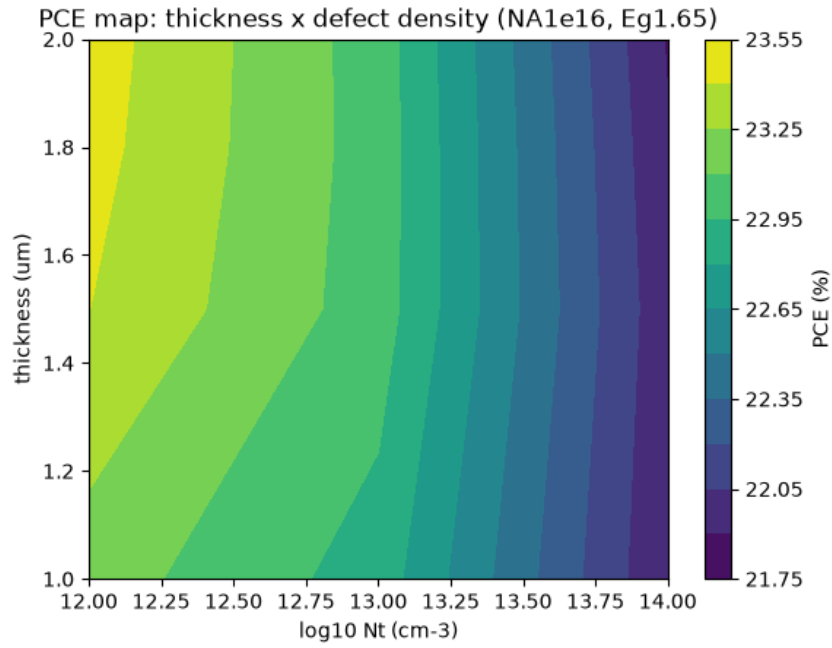


Figure 3: Efficiency map over absorber thickness and bulk defect density (slice at $N_A = 10^{16} \text{ cm}^{-3}$, $E_g = 1.65 \text{ eV}$).

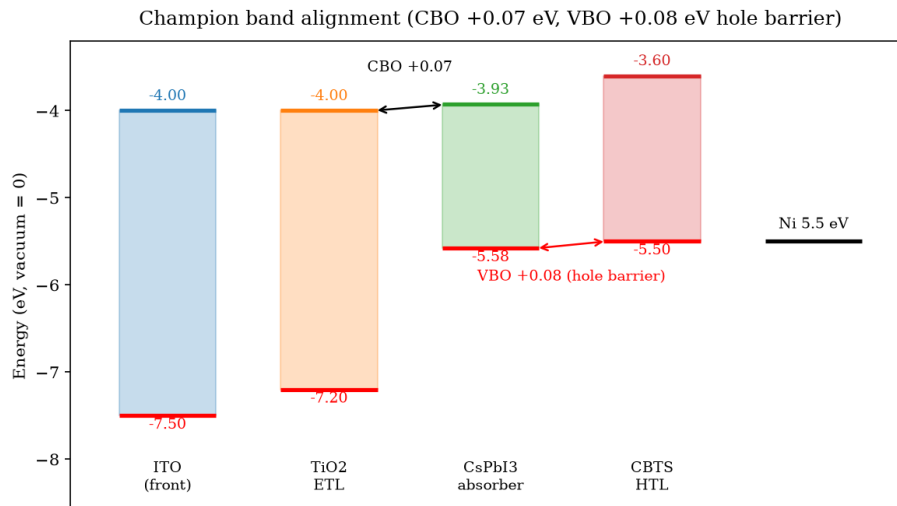


Figure 4: Equilibrium band alignment of the champion stack from input parameters (energies in eV vs vacuum).

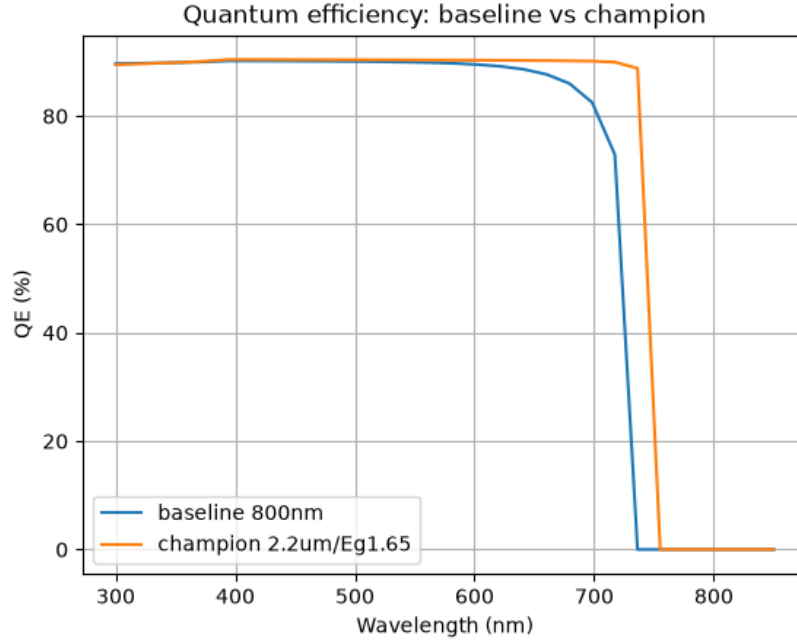


Figure 5: Quantum efficiency of baseline versus champion; the edge shifts 731 to 752 nm.

Jsc gain. The UV plateau assumes the definition file’s flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO₂, so it is an upper envelope in the blue rather than a prediction. Capacitance–voltage analysis at 1 MHz (Figure 6) shows rising capacitance with forward bias and a linear Mott–Schottky branch; the extracted depletion width narrows from 0.29 μm at 0 V to 0.22 μm at 0.5 V, an order of magnitude below the 2.2 μm absorber thickness throughout. The illuminated band diagram at short circuit (Figure 7) exhibits clean quasi-Fermi splitting, and generation exceeds recombination by at least two orders of magnitude at every mesh point of the full 2.33 μm profile (minimum ratio $\sim 3 \times 10^2$, rising to $\sim 10^4$ – 10^5 near the back contact; Figure 8).

Temperature dependence (Figure 9, Table 7) follows the established trend [5]: efficiency falls from 24.44% at 275 K to 16.70% at 475 K, driven by Voc (1.271 to 0.955 V) and FF while Jsc stays pinned at $\sim 21.19 \text{ mA/cm}^2$, i.e. $-0.16\%/K$ relative (-0.039 points/K) over 300–400 K. The 275 K fill factor (90.75%) sits *below* Green’s ideal-diode value at that Voc and temperature (90.86%), so no artefact needs invoking; likewise the champion’s 89.78% is within 0.2 points of ideal (90.0%), leaving essentially no FF headroom beyond series resistance. The 475 K end of the sweep ignores the temperature dependence of E_g and black-phase stability, so it is a solver stress test, not a prediction.

6.1 Parasitic-resistance accounting

Series and shunt resistances are scriptable in SCAPS 3.3.10 after all (`set external.Rs/Rsh`; an earlier draft incorrectly claimed otherwise and used an analytic derate). Measured scripted sweeps on the champion give 23.29% at $R_s = 0.5 \Omega\text{cm}^2$, **23.07%** at $R_s = 1 \Omega\text{cm}^2$, and 22.64% at $R_s = 2 \Omega\text{cm}^2$ (Table 8), agreeing with the analytic derate to 0.02 points. Shunt paths cost 0.11 points at $R_{sh} = 10^4 \Omega\text{cm}^2$ (23.38%) and 1.27 points at $R_{sh} = 10^3 \Omega\text{cm}^2$ (22.22%); combined $R_s = 1/R_{sh} = 10^4$ gives 22.95%. The upper bound therefore survives realistic parasitics, but only just: at $R_s = 2 \Omega\text{cm}^2$ the margin over the 22.02% certified record is 0.6 points.

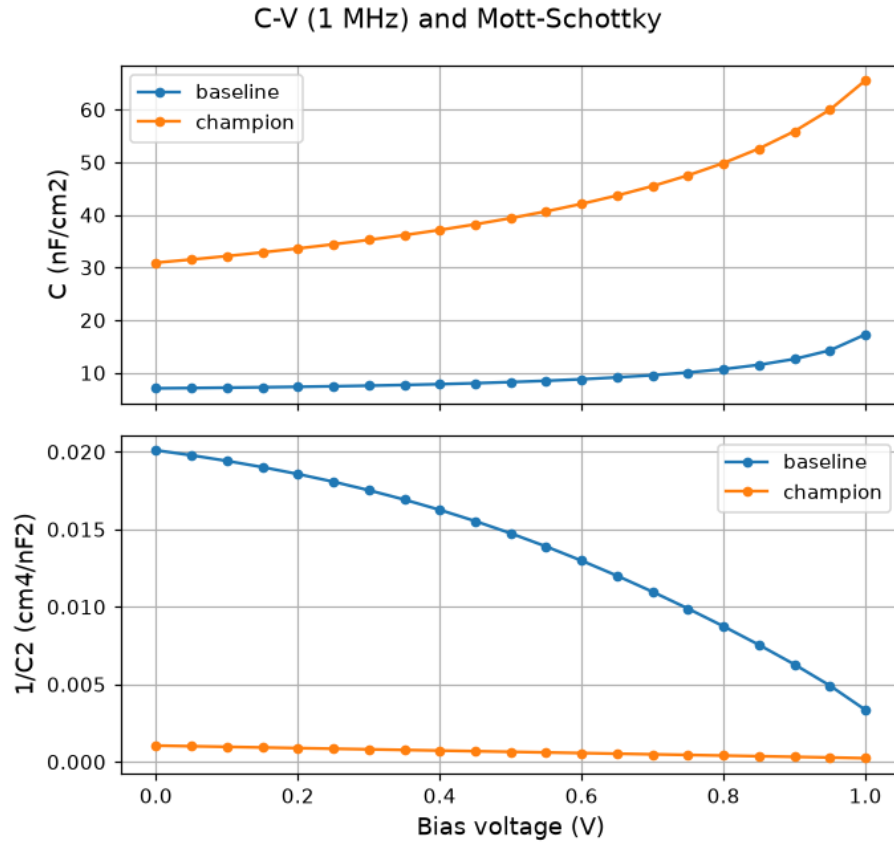


Figure 6: Capacitance–voltage (1 MHz) and Mott–Schottky plots for baseline and champion.

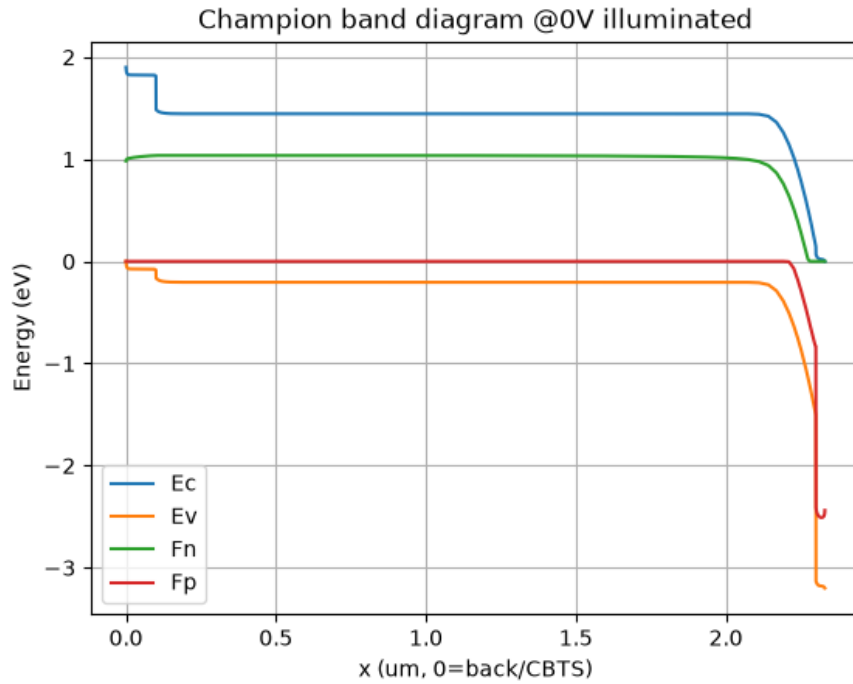


Figure 7: Illuminated band diagram of the champion at short circuit ($x=0$ at the back contact).

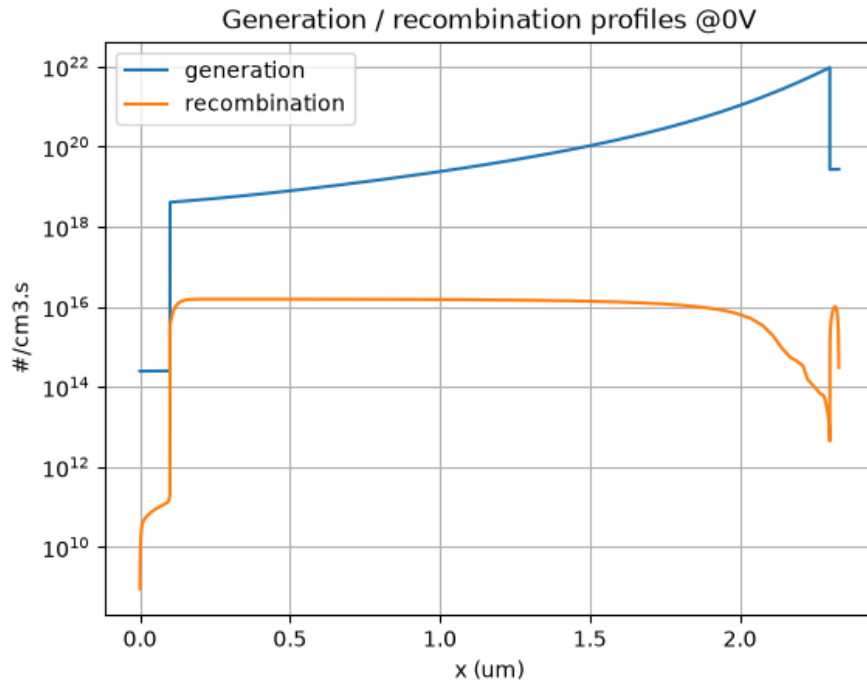


Figure 8: Generation and recombination profiles at short circuit (log scale).

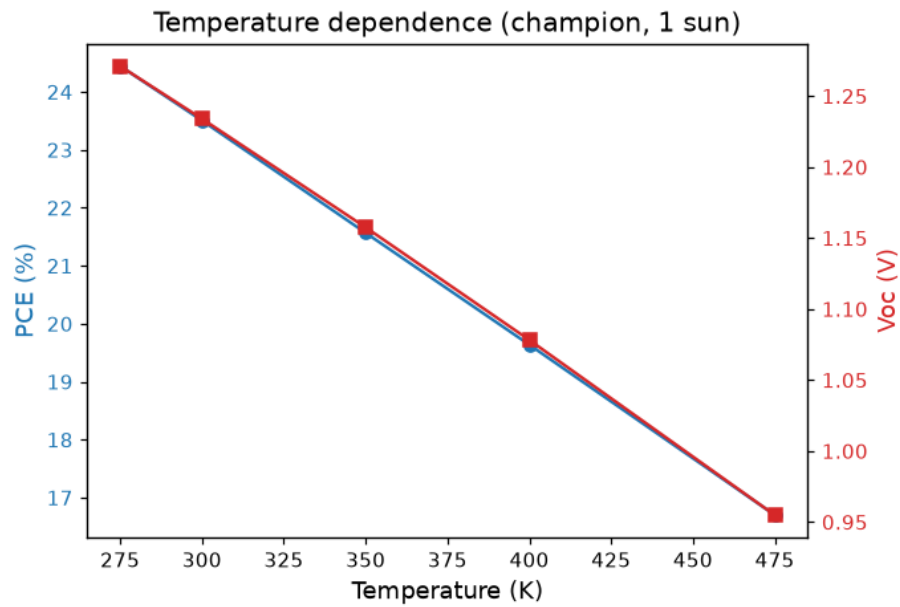


Figure 9: Efficiency and open-circuit voltage versus operating temperature (champion).

Table 7: Temperature dependence of the champion (1 sun). The 300 K row uses the fine-IV-step confirmatory run (23.49%); all other rows use coarse-grid sweep values.

T (K)	Voc (V)	Jsc (mA/cm ²)	FF (%)	PCE (%)
275	1.271	21.19	90.75	24.44
300	1.234	21.19	89.78	23.49
350	1.157	21.20	87.91	21.57
400	1.078	21.20	85.92	19.64
475	0.955	21.20	82.50	16.70

Table 8: Champion under measured scripted parasitics (all values computed, none derated).

Parasitics	PCE (%)	vs 22.02% cert.	Status
$R_s = 0$ (upper bound)	23.49	+1.47	SCAPS upper bound
$R_s = 0.5 \Omega\text{cm}^2$	23.29	+1.27	realistic good contact
$R_s = 1 \Omega\text{cm}^2$	23.07	+1.05	realistic typical
$R_s = 2 \Omega\text{cm}^2$	22.64	+0.62	margin nearly gone
$R_{sh} = 10^4 \Omega\text{cm}^2$	23.38	+1.36	mild shunt
$R_{sh} = 10^3 \Omega\text{cm}^2$	22.22	+0.20	leaky cell
$R_s = 1 + R_{sh} = 10^4$	22.95	+0.93	combined realistic

7.0 Reproducibility audit: what the 24%+ claims are made of

Deleting the interface blocks inflates efficiency by 1.26–1.67 points absolute: 23.40 to 24.87%, 23.47 to 24.73%, and 23.50% coarse (23.49% fine) to 25.16% (Figure 10), almost entirely through a ~ 0.13 V Voc rise as interface recombination vanishes. But the binary ON/OFF comparison understates the audit: the full dose–response (Table 9) runs 25.04% at 10^8 cm^{-2} , 24.54% at 10^9 cm^{-2} , 23.49% at the 10^{10} cm^{-2} guardrail, 22.88% at $3 \times 10^{10} \text{ cm}^{-2}$, 22.21% at 10^{11} cm^{-2} , and 20.64% at 10^{12} cm^{-2} . The published 24.17% [7] and 24.24% [8] values fall *below* our no-interface band (24.73–25.16%) and land between the 10^8 and 10^9 cm^{-2} levels of our curve—consistent with interface quality roughly $10\times$ below our guardrail, not with fully omitted interfaces. We could not re-run either study from its published parameters (neither paper prints a complete parameter table), so this placement is inference from our dose curve, stated as such. Under interface-inclusive settings, the highest reported CsPbI3-CBTs value we can verify remains 23.10% [21]; our 23.49% is the highest *guardrailed* value on this stack only under the assumptions of Section 3.0, and Section 9.0 itemises what could still deflate it.

7.1 Where the missing points go

The inflation is almost entirely a Voc effect (+0.13 V) with Jsc unchanged to within 0.02 mA/cm^2 : removing the 10^{10} cm^{-2} interface states eliminates the dominant SRH channel at the heterojunctions, unpinning the quasi-Fermi splitting toward the radiative limit—indeed the no-interface Voc values (1.363–1.382 V) sit essentially *at* the ~ 1.37 V radiative limit for 1.65 eV, which is itself evidence that the interface-inclusive numbers, not the inflated ones, live in the physical regime. Fill factor moves non-monotonically (89.8 to 87.0% on the champion) as the recombination profile reshapes. Asymmetric tests show both heterojunctions matter: 10^{12} cm^{-2} on the ETL side alone gives 21.35%, on the HTL side alone 21.20%. Raising the capture cross-section $10\times$ at fixed $N_{if} = 10^{10} \text{ cm}^{-2}$ costs 0.63 points (23.86%), and bulk $N_t \times \sigma$ interacts: $\sigma \times 10$ costs 0.23 points at the 10^{12} cm^{-3} floor but 1.5 points at routine 10^{15} cm^{-3} (17.44%). The measured parasitic sweep applied to the guarded champion is tabulated below.

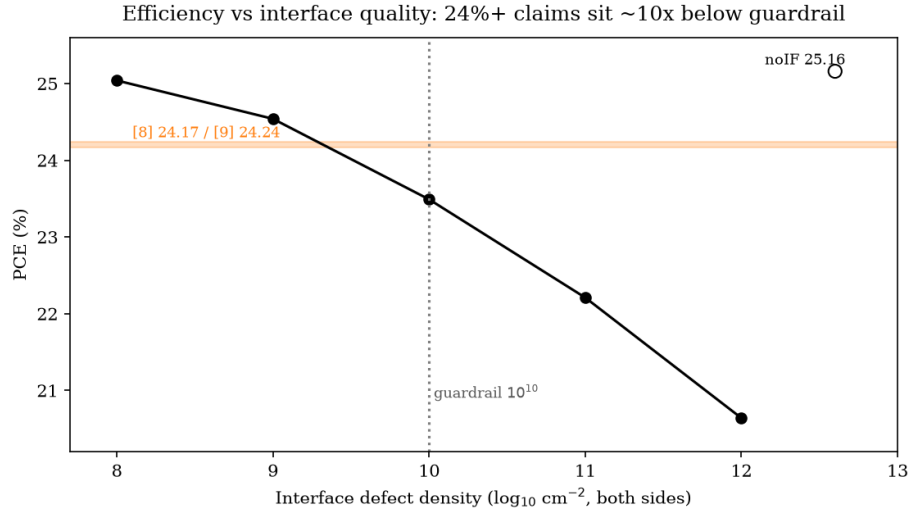


Figure 10: Same cell at graded interface quality: efficiency vs N_{if} (both sides) with the 24%+ literature band and the no-interface point. The claims sit $\sim 10\times$ below guardrail, not at zero interfaces.

Table 9: Interface dose–response on the champion (both sides unless noted).

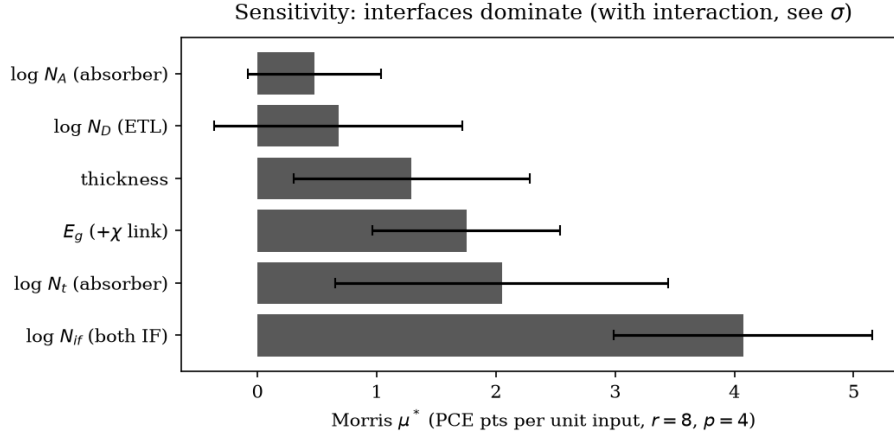
$N_{if} (\text{cm}^{-2})$	PCE (%)	Note
10^8	25.04	above all literature
10^9	24.54	above 24.17/24.24 claims
10^{10} (guardrail)	23.49	this work
3×10^{10}	22.88	below certified record pace
10^{11}	22.21	–
10^{12}	20.64	–
10^{12} ETL side only	21.35	HTL side guarded
10^{12} HTL side only	21.20	ETL side guarded
10^{10} , $\sigma \times 10$	23.86	cross-section sensitivity
removed (noIF)	25.16	at radiative Voc limit

Table 10: Guardrail audit: identical cells with interfaces ON versus OFF.

Cell	IF ON (%)	IF OFF (%)	Inflation (pts)
1.5 μm cell	23.40	24.87	+1.47
2.0 μm cell	23.47	24.73	+1.26
Champion	23.49	25.16	+1.67

Table 11: Morris screen design and results ($r = 8$ trajectories, $p = 4$, $\Delta = 2/3$, seed 7; 56/56 converged). μ^* in PCE points per unit normalised input.

Factor	Range	μ^*	σ	n
$\log N_{if}$ (both IF)	10^8 – 10^{12} cm^{-2}	4.08	1.08	16
$\log N_t$ (absorber)	10^{12} – 10^{15} cm^{-3}	2.05	1.40	16
E_g (+ χ link)	1.65–1.75 eV	1.75	0.79	16
Thickness	0.4–2.2 μm	1.29	0.99	16
$\log N_{D,ETL}$	10^{16} – 10^{19} cm^{-3}	0.68	1.04	16
$\log N_A$ (absorber)	10^{15} – 3×10^{16} cm^{-3}	0.48	0.56	16

**Figure 11:** Morris μ^* (bars) with interaction spread σ (whiskers). Interfaces dominate; large σ throughout signals interactions.

8.0 Morris sensitivity screening

An earlier draft ranked factors with a random forest trained on the grid runs; that analysis is withdrawn, because importance scores from a coarse grid with uneven ranges measure the sampling design, not the physics. We instead ran a Morris sensitivity screen: $r = 8$ trajectories, $p = 4$ levels, $k = 6$ factors (absorber thickness, $\log N_A$, $\log N_t$, $\log N_{D,ETL}$, E_g with the χ link applied deterministically, and $\log N_{if}$ tied on both interfaces), 56 fresh simulations, all converged, spanning 15.6–23.4% (Table 11, Figure 11). Interface quality dominates ($\mu^* = 4.08$ PCE points per unit input), ahead of bulk defect density (2.05) and bandgap (1.75), with thickness (1.29), ETL doping (0.68), and absorber doping (0.48) trailing. The large σ values throughout confirm strong interactions/nonlinearity—notably thickness acting through defect-limited diffusion length—so μ^* is a screening rank, not a variance share. Two disclosures: E_g moves together with χ by construction, and the two interfaces move together, so their scores bundle linked effects. The substantive correction stands regardless: on this stack, interfaces outrank bulk defects, reversing the withdrawn ranking.

The ranking agrees with Bayesian optimisation on CsSnI3, where absorber defect density likewise dominates (mean $|\text{SHAP}| = 3.21$ versus 1.67 for thickness) and the optimal thickness shifts with N_t [9]. Surface-passivated CsPbI3 cells confirm that interface-adjacent defects gate the voltage [29], and the 7182-configuration open dataset now makes such cross-study surrogate comparisons routine [30].

9.0 Discussion

Fabrication plausibility bounds the result, and new sensitivity runs tighten the bounds. $N_A = 3 \times 10^{16} \text{ cm}^{-3}$ is routinely achievable; $E_g = 1.65 \text{ eV}$ sits below the $\sim 1.7 \text{ eV}$ of the experimental records and contributes roughly one point (best 1.694 eV grid point: 22.51%); $N_t = 10^{12} \text{ cm}^{-3}$ demands near-single-crystal quality, and raising the capture cross-section $10\times$ costs a further 0.23 points there and 1.5 points at routine 10^{15} cm^{-3} . The contacts are not neutral choices: lowering Ni from 5.5 to 5.0 eV (the usual metal range) costs 0.23 points, and raising ITO from 4.0 to 4.7 eV collapses the cell to 18.60% (Voc 1.08 V), so the front contact must stay near 4.0–4.4 eV. Together with the +0.08 eV hole barrier of Section 5.0, the 23.49% is an upper bound contingent on near-ideal contacts as well as near-ideal bulk. Against experiment: 23.49% exceeds the 22.02% certified record by 1.47 points and the 22.60% laboratory value by 0.89 points; with measured $R_s = 1 \Omega\text{cm}^2$ (23.07%) the margin over certified is 1.05 points and over laboratory 0.47 points. The result sits at $\sim 78\%$ of the $\sim 30\%$ Shockley–Queisser limit at 1.65 eV ($\sim 1.45 \text{ V}$), well inside thermodynamics, unlike the explicitly idealised 40.17% MASnI3 study [24]. A 40% single-junction device at one sun remains out of reach for any lead-halide gap; the credible path past 30% is a CsPbI3/Si or all-perovskite tandem using the 1.65 eV recipe as the top cell.

10.0 Outlook

A laboratory translation would need, simultaneously: $2.0\text{--}2.2 \mu\text{m}$ CsPbI3 films at single-crystal-grade defect levels (time-resolved photoluminescence verified), interface passivation toward 10^{10} cm^{-2} following the fluorinated-dipole [2] or dual-interface [3] chemistries, a front contact near 4.0–4.4 eV, and $R_s \leq 1 \Omega\text{cm}^2$ (Table 8). We state plainly what is not demonstrated: a $2.2 \mu\text{m}$ device-quality CsPbI3 film at $N_t \sim 10^{12} \text{ cm}^{-3}$, and sputtered CBTS directly on CsPbI3 (CBTS typically requires high-temperature sulfurisation), for which we found no precedent—so this section is a requirements list, not a recipe. The near-term fab target from our own map is $\sim 19\%$ at sub-micron thickness and routine defects. In-line controls, where applicable: QE edge at 752 nm (Figure 5) and the $-0.16\%/K$ temperature coefficient (Table 7).

11.0 Limitations and outlook

Five boundaries delimit this work. First, $N_t = 10^{12} \text{ cm}^{-3}$ at $2.2 \mu\text{m}$ demands near-single-crystal films; at the routinely achieved 10^{15} cm^{-3} our own map (Figure 3) puts the optimum near 19% at sub-micron thickness, which is the near-term fab target. Second, the champion's +0.08 eV hole barrier and contact work functions (Ni 5.5 eV, ITO 4.0 eV) are tuned, not standard—Section 9.0 quantifies each penalty. Third, phase stability of black CsPbI3 against the yellow δ -phase, ion migration, $E_g(T)$, and damp-heat degradation are outside SCAPS-1D steady-state scope and must be validated experimentally, following the passivation routes of [1–3]. Fourth, the grid is a screen, not an optimiser: the optimum sits at the corner of the allowed box and is numerically indistinguishable from its neighbours. Fifth, the single-junction ceiling ($\sim 30\%$ at 1.65 eV [25]) caps how far any interface-inclusive optimisation can go; the credible path past 30% is a CsPbI3/Si or all-perovskite tandem using the 1.65 eV recipe as the top cell. The lead-free programme can inherit the guardrailed screening protocol directly.

12.0 Conclusion

Grid screening under interface-inclusive guardrails delivers a 23.49%-efficient CsPbI3/CBTS upper bound whose every assumption is priced: single-crystal-grade defects, a low gap, tuned contacts, and a small hole barrier. The same campaign shows the realistic-defect result is $\sim 19\%$, measures

the interface dose–response that re-places the published 24%+ values, replaces an analytic resistance derate with scripted parasitic sweeps, and ranks interfaces first by Morris screening. We recommend guardrailed reporting (interface status *with* dose level, defect floors *with* cross-sections, unit-system declaration, Rs basis, contact work functions) as a publication standard for SCAPS studies.

Data availability

All definition files (csPbI₃-CBTS-r7.def with interfaces, csPbI₃-CBTS-noIF.def without, plus work-function and cross-section variants), sweep receipts (roundA.jsonl, roundB.jsonl, char.jsonl, s0_verify.jsonl, s1_dose.jsonl, s4_sig.jsonl, s5_morris.jsonl), raw curves (champion_JV.jsonl), analysis outputs and figure scripts are archived with the project under leadpaper/ (inventory in Table 12; directories defs/, outputs/, receipts/, logs/). A public repository with DOI will be released on acceptance; until then all files are available from the authors.

Author contributions

The authors designed the screening study, built the SCAPS definition files, executed the simulation campaign, performed the audit and sensitivity analyses, and wrote the manuscript.

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Appendix A: the scripting unit trap

SCAPS definition files store SI units (metres, m⁻³, m²/Vs) but script `set` commands expect practical units (μm, cm⁻³, cm²/Vs)—verified by dose–response: `set layer2.NA 1e15` reproduces the file baseline (stored 10²¹ m⁻³) exactly, while SI-valued inputs (e.g. 10²²) inject a 10⁶× overdose and yield degenerate ghost cells (Voc 1.47–1.53 V, FF 91–92%, up to 27%). Practical flags that catch them: Voc above the radiative limit for the modelled gap, or FF above Green’s ideal-diode value at the measured Voc. All values in this paper are stated in the units the script layer expects; the archived logs record both.

Appendix B: run inventory

All simulation batches are inventoried in Table 12.

Table 12: Simulation batches archived under leadpaper/.

Batch	Configurations	Receipt
Baseline validation	4 (one worker-crash retry)	baseline_results.json
Unit diagnostics	6 + 4 retries	diag_units.json
Round A grid	96 (3 non-converged)	roundA.jsonl
Round B refine	36 (3 non-converged)	roundB.jsonl
Characterisation + audit + T	17 + 6 re-runs	char.jsonl
Set-command verification	5	s0_verify.json
Interface dose + Rs/Rsh + WF	14	s1_dose.json
$N_t \times \sigma$ joint	3	s4_sig.json + s4b_sig.json
Morris screen	56 (0 errors)	s5_morris.jsonl